SYNTHONICON ALGEBRAIC OPERATIONS — TENSOR-MATH EDITION meet() · join() · tensor(⊗)· lift · path · monad · decompositions

ğ1 MEET () — Greatest Lower Bound

1.1 meet(Hv1_human_open, AtHv1_primed)

SETUP: Hv1_human_open : $\langle D_\wedge; T_\bowtie; R_\supseteq; P_{+-}; F_\hbar; K_{\mathrm{mod}}; G_\sqsupset; \Gamma_\wedge(\mathrm{SEL}); \Phi_c; \Omega_0\rangle$ AtHv1_primed : $\langle D_\wedge; T_\bowtie; R_\supseteq; P_{+-}; F_\hbar; K_{\mathrm{mod}}; G_\sqsupset; \Gamma_\wedge(\mathrm{SEL}); \Phi_{\mathrm{sub}}; \Omega_0\rangle$

TENSOR-MATH ANALOGY: In a product of bounded lattices $L_1 \times L_2 \times \cdots \times L_n$, the meet is computed component-wise. Here Φ-space is a 3-element lattice $\Phi_{\mathrm{sub}} <? < \Phi_c$ where Φ_c is the TOP element (absorbing). So ⊓ on Φ returns Φ_c regardless of which operand holds it. This is NOT the usual infimum — it is an absorbing element, analogous to the "⊤ element in a co-Heyting algebra."

[RESULT] STATUS : PASS $\implies$ Φ: $\Phi_c \sqcap \Phi_{\mathrm{sub}} \to \Phi_c$ (Φ_c dominates in meet)

INSIGHT: $d(\mathrm{AtHv1\_primed}, \mathrm{PsHv1\_constitutive}) = 0.000$ — the gymnosperm channel is algebraically identical to mechanically primed *Arabidopsis*. The meet is the algebraic proof of cross-species functional conservation.

1.2 meet(Hv1_human_open, 2GBI_inhibitor) — correct conflict

SETUP: Hv1_human_open : $T_\bowtie$ (H-bond network topology, cyclic) 2GBI_inhibitor : $T_\in$ (condensed bicyclic, rigid ring network) + $P^\psi_\pm$ (pseudosymmetric) vs P_{+-} (directional)

TENSOR-MATH ANALOGY: In module theory: two modules M, N have a non-trivial meet (intersection) only when they are sub-objects of a common ambient module. $T_\bowtie$ and $T_\in$ are NOT in the same T-equivalence class — there is no common parent topology. The meet is ⊥: undefined.

[RESULT] STATUS : CONFLICT T P $\implies$ Φ: $\Phi_c \sqcap \Phi_{\mathrm{sub}} \to \Phi_c$ (Φ_c dominates in meet) $\implies$ F: $F_\hbar \sqcap F_{\mathrm{eth}} \to F_{\mathrm{eth}}$ (min)

INSIGHT: The conflict is the answer. 2GBI occludes the channel — it does NOT adopt the channel's topology or merge with it. The drug sits in the pore and blocks it physically. tensor() is the correct operation for that question (see ğ3). meet() correctly returns ⊥.

1.3 Topological protection order: meet(Cooper pair, TI surface)

SETUP: cooper_pair : Ω_Z (ℤ winding number; BCS s-wave) topological_insulator : Ω_{Z_2} ($\mathbb{Z}_2$ time-reversal; Bi_2Se_3 class)

Ω ORDINAL: TRIVIAL(0) < Z2(1) < Z(2) < CHERN(3) < NON_ABELIAN(4)

TENSOR-MATH ANALOGY: The Ω lattice mirrors the Altland-Zirnbauer (AZ) classification. Under meet, Ω behaves like a meet-semilattice: the conservative guarantee is the WEAKER protection class — you can guarantee the physics at the intersection only up to the least protected invariant. This is the lattice-theoretic reason topological surface states are fragile to symmetry-breaking perturbations that act on the weaker $\mathbb{Z}_2$ invariant.

NOTE: This is a topology-focused meet. D and T will conflict here (MOLECULAR vs SUPRAMOLECULAR, BOWTIE vs BOWTIE — actually BOWTIE matches). The pedagogical point is the Ω ordering rule.

[RESULT] STATUS : CONFLICT D P Γ $\implies$ Φ: $\Phi_c \sqcap \Phi_{\text{sub}} \to \Phi_c$ (Φ_c dominates in meet) $\implies$ Ω: $\Omega_Z \sqcap \Omega_{Z_2} \to \Omega_{Z_2}$ (min protection) $\implies$ K: $K_{\text{slow}} \sqcap K_{\text{fast}} \to K_{\text{slow}}$ (min)

ğ2 JOIN () — Least Upper Bound / Design Target

2.1 join(Hv1_human_open, PsHv1_constitutive)

SETUP: Hv1_human_open : Φ_c (H-bond water chain) PsHv1_constitutive : Φ_{sub} (constitutively primed; no Φ_c required)

TENSOR-MATH ANALOGY: In a module category, join is the pushout: the minimal object receiving morphisms from both M and N. Here the join forces Φ_c into the design target — the coproduct must accommodate the more demanding constraint (criticality) from the human channel.

[RESULT] STATUS : PASS $\implies$ Φ: join propagates Φ_c (criticality is join-dominant)

2.2 join(2GBI_inhibitor, HIF_inhibitor) — no common design target

SETUP: 2GBI : $T_\in$ (condensed fused bicyclic: charge-delocalised ring system) HIF : $T_|$ (flexible linear linker: two pharmacophores, conformational freedom)

TENSOR-MATH ANALOGY: These are objects from DIFFERENT categories: $T_\in$ is in the "network" homotopy class; $T_|$ is in the "linear" class. The coproduct (join) requires a common ambient object — but no scaffold topology subsumes both rigid bicyclic AND flexible linear in the same T class. The join is undefined ($\perp$ in the join-semilattice for T).

DESIGN CONSEQUENCE: This algebraically forbids "best of both worlds" scaffold merging. The correct design path is tensor() (ğ3.3): what do they predict TOGETHER as a co-assembly, not as a single molecule?

[RESULT] STATUS : CONFLICT T

2.3 join(imatinib, GNF2) — combined Type II + allosteric target

SETUP: imatinib : K_{slow}, G_{local}, $\Gamma_\wedge$(SPECIFIC), Φ_{sub} GNF-2 : K_{mod}, $G_\gimel$, $\Gamma_\to$(SELECTIVE), Φ_c

TENSOR-MATH ANALOGY: join on ordered dimensions takes supremum: $\max(K_{\text{slow}}, K_{\text{mod}}) = K_{\text{slow}}$ (the slowest kinetics must be satisfied by the design target). G: $\max(G_\beth, G_\gimel) = G_\gimel$ (target must have mesoscale propagation). Φ_c propagates. The join is the "hardest" requirement from either source — a design constraint satisfaction problem whose solution is the join element.

Note: T conflict expected ($T_\bowtie$ vs T_{branched}). This is the real bottleneck: combining the DFG-out loop topology with the myristoyl pocket branched topology requires T-class redesign.

[RESULT] STATUS : CONFLICT T Γ $\implies$ Φ: join propagates Φ_c (criticality is join-dominant) $\implies$ F: $F_\hbar \sqcup F_{\text{eth}} \to F_\hbar$ (max) $\implies$ K: $K_{\text{slow}} \sqcup K_{\text{mod}} \to K_{\text{mod}}$ (max) $\implies$ G: $G_\beth \sqcup G_\gimel \to G_\gimel$ (max)

INSIGHT: The T-conflict reveals the hard part of dual-mechanism ABL inhibitor design: the DFG-out pocket ($T_\bowtie$, cyclic coordination) and the myristoyl pocket (T_{branched}, pendant ligand) cannot be joined without a new T topology. This correctly predicts why Type II + allosteric dual-binders have been rare and difficult to design.

ğ3 TENSOR (⊗) $---Bifunctor/Co-AssemblyPrediction$

3.1 tensor(electron, hole) — exciton precursor; bound state gap

SETUP: electron : D_{mol}, $T_{|}$, P_{donor}, F_{high}, K_{fast}, G_{local}, Ω_0 hole : D_{mol}, $T_{|}$, P_{acceptor}, F_{high}, K_{fast}, G_{local}, Ω_0

KEY PREDICTION: $T_{|} \otimes T_{|} \to T_{|}$ (no topology promotion for SAME input)

TENSOR-MATH ANALOGY: In Hilbert space: $\mathcal{H}_e \otimes \mathcal{H}_h$ gives the two-particle space. The Coulomb interaction H_{Coulomb} is an element of $\mathrm{End}(\mathcal{H}_e \otimes \mathcal{H}_h)$ — it is NOT included in the tensor product itself. The bound exciton lives in a subspace selected by H_{Coulomb}, which requires a separate 'meet with binding potential' operation in synthon algebra.

This example teaches the semantic boundary: tensor() = statistical co-occupancy (two-particle Hilbert space) meet() = intersection (bound-state subspace selection) The exciton = tensor(e, h) >> meet(coulomb_binding_potential_synthon)

[RESULT] STATUS : PASS $\implies$ P: $P_{\mathrm{minus}} \otimes P_{\mathrm{plus}} \to P_{\mathrm{pm\_pseudo}}$ (charge pairing) $\implies$ ξ_{CP}: $7.741 + 7.741 - 0.5 \times 6.635 = 12.164$ nats (6/7 primitives match $\to I \approx 6.635$ nat) ξ_{CP} : 12.164 nats

INSIGHT ($P \to$ DONOR_ACCEPTOR): P: DONOR ⊗ ACCEPTOR → DONOR_ACCEPTOR is the primordial signature of charge-transfer: electron and hole have opposing charge asymmetry and the tensor correctly predicts a directed dipole will form. Frenkel exciton (T stays LINEAR) vs Wannier-Mott ($T_{\bowtie}$ via binding) distinguished by whether the binding step is allowed.

3.2 tensor(phonon_acoustic, magnon, $\lambda = 0.4$) — magnetoelastic polaron

SETUP: phonon_acoustic : D_{supra}, $T_{|}$, P_{sym}, F_{low}, K_{fast}, G_{global}, Ω_0 magnon : D_{supra}, $T_{|}$, P_{sym}, F_{med}, K_{mod}, G_{meso}, Ω_0

COUPLING: $\lambda = 0.4$ (moderate spin-phonon coupling, e.g. YIG or MnF_2)

TENSOR-MATH ANALOGY: The magnon-phonon Hamiltonian $H_{\mathrm{mp}} = \sum g_{kq}(a_k + a_k^\dagger)(b_q + b_q^\dagger)$ is a bilinear coupling in the product Fock space. In the synthon algebra this is exactly tensor(): the co-assembly of two SUPRA collective modes. The λ parameter encodes g_{kq} coupling strength — high λ reduces ξ_{tensor} via mutual information discount.

EXPECTED: $G \to G_{\mathrm{global}}$ (phonon bath extends over whole crystal); $F \to F_{\mathrm{low}}$ (phonon decoherence is the bottleneck); $K \to K_{\mathrm{fast}}$ (acoustic phonon velocity dominates).

[RESULT] STATUS : PASS $\implies$ F: $\min(F_\ell, F_{\mathrm{eth}}) \to F_\ell$ (bottleneck) $\implies$ K: $\min(K_{\mathrm{fast}}, K_{\mathrm{mod}}) \to K_{\mathrm{mod}}$ (trap risk propagates) $\implies$ G: $\max(G_\aleph, G_\beth) \to G_\aleph$ (coarsest scale dominates) $\implies$ Γ: $\Gamma_{\mathrm{or}} \otimes \Gamma_{\mathrm{and}} \to \Gamma_\wedge$(BROAD) (AND-composition) $\implies$ ξ_{CP}: $7.507 + 7.590 - 0.4 \times 4.290 = 13.381$ nats (4/7 primitives match $\to I \approx 4.290$ nat) ξ_{CP} : 13.381 nats

3.3 tensor(Majorana, Majorana, $\lambda = 0.3$) — topological qubit

SETUP: majorana_fermion : D_{mol}, T_{braid}, P_{sym}(self-conjugate), F_{high}, K_{trap}, G_{local}, Φ_c, Ω_{NA}

SPECIAL RULE: $T_{\mathrm{braid}} \otimes T_{\mathrm{braid}} \to T_{\mathrm{braid}}$ (braided exchange statistics are preserved in co-assembly; anyonic topology does NOT network-promote like spatial structures)

TENSOR-MATH ANALOGY: The braid group B_n has a natural tensor structure: $B_n \otimes B_m \subseteq B_{n+m}$ as a sub-group. The braid topology of two Majorana modes tensors to give a larger braid system, not a "network" of modes. This is why T_{braid} has its own special tensor rule: it does not obey the standard topology promotion hierarchy.

Ω rule: $\Omega_{\text{NA}} \otimes \Omega_{\text{NA}} \to \Omega_{\text{NA}}$ (non-Abelian protection preserved)

PHYSICAL MEANING: Two spatially separated Majorana modes in a Kitaev chain form a topological qubit with Ω_{NA} protection. The tensor product predicts this ensemble correctly: T_{braid} preserved, Ω_{NA} preserved, G: $\max(\text{LOCAL}, \text{LOCAL}) = \text{LOCAL}$ (both modes are localised).

[RESULT] STATUS : PASS $\implies T$: $T_{\text{braid}} \otimes T_{\text{braid}} \to T_{\text{braid}}$ (anyonic statistics preserved under composition) $\implies \xi_{\text{CP}}$: $8.314 + 8.314 - 0.3 \times 8.314 = 14.134$ nats (7/7 primitives match $\to I \approx 8.314$ nat) ξ_{CP} : 14.134 nats

CONTRAST WITH EXAMPLE 3.1: electron $\otimes hole$:$\text{T}_{|} \otimes T_{|} \to T_{|}$ (same topology, no promotion) majorana $\otimes majorana$:$\text{T}_{\text{braid}} \otimes T_{\text{braid}} \to T_{\text{braid}}$ (braid preserved) magnon $\otimes phonon$:$\text{T}_{|} \otimes T_{|} \to T_{|}$ (same topology, no promotion)

```
The ONLY cases where tensor changes $T$ are cross-topology
   products:
$T_{|} \otimes T_{\in} \to T_{\in}$  (linear + network ->
   network)
$T_{\in} \otimes T_{\Box} \to T_{\Box}$  (network + cage ->
   cage)
This mirrors the intuition: co-assembly of a network and a
   linear
entity produces network-class organisation, not vice versa.
```

ğ4 LIFT — Natural Transformations Between Domain Categories

4.1 lift_to_temporal — molecule enters catalytic cycle category

RULES: $D_{\wedge} \to D_{\infty}$ (point object $\to$ temporal / periodic) $T \to T_{\bowtie}$ (any $\to$ bowtie: substrate-in / product-out loop) $R_{\supseteq} \to R_{\ddagger}$ (non-covalent $\to$ dynamic catalytic / turnover) $K_{\text{fast}} \to K_{\text{mod}}$ (fast-exchange adjusted for catalytic dwell time) $\Gamma \to \Gamma_{\to}$(SEL) (sequential: binding-order enforced)

TENSOR-MATH ANALOGY: In category theory: D_{∞} objects are "internal monoids" — objects equipped with a self-map (the catalytic step). The temporal lift is the functor that sends a morphism (binding event) to a monoid action (the catalytic cycle). It is the "loop-space" functor Ω in topology: $\Omega(X)$ maps a pointed space to its loop space, endowing it with a group structure. Here the "group" is the catalytic turnover.

COST: 0.0 nats (no thermodynamic cost to describe the catalytic frame) GROUNDING REQUIRED: Axiom 6 — must specify a reset event (product release, cofactor regeneration). lift alone does not ground; grounding is a separate obligation enforced by the Axiom validator.

Found temporal synthon in catalog: global_supply_chain Notation: $\langle D_{\infty}; T_{\text{network}}[1 : *]; R_{\ddagger}; P_{\text{minus}}; F_{\text{eth}}; K_{\text{mod}}; G_{\aleph}; \Gamma_{\text{and}}(\text{SELECTIVE}); \Phi_{\text{sub}}; 1 : *\rangle$

4.2 lift_to_spatial — molecule → crystal secondary building unit

RULES: $D_\wedge \to D_\triangle$ (molecular → supramolecular crystal) $T \to T_\square$ (any → hub-node: SBU topology) $G_\beth \to G_\daleth$ (min) (local → mesoscale; crystal requires mesoscale) $\Gamma \to \Gamma_\wedge$(SEL) (coordinated multi-dentate recognition)

TENSOR-MATH ANALOGY: The spatial lift is the ”classifying space” functor B: $B(G)$ takes a group G (the molecule's local symmetry) and produces a space whose loop space is G. In crystal engineering terms: the SBU is the ”classifying object” for the crystal packing symmetry. The $T_\square$ (hub) topology encodes the branching valence of the SBU — how many struts radiate from the node.

A PRACTICAL CONSEQUENCE: lift_to_spatial(2GBI_inhibitor) would give $T_\square$ — but 2GBI is bicyclic ($T_\in$). The T change ($\in \to \square$) has a real cost in synthesis: you must redesign the ring to have branching coordination points. This is not a free transformation.

4.3 criticality_lift — non-zero cost functor, fidelity gate

GATE: $F \geq F_\hbar$ required to lift $\Phi_{\text{sub}} \to \Phi_c$ COST: +2.303 nats (one decade of probability: $\log_e(10)$)

This is the primitive-space analog of the LANDAUER BOUND: In information theory: erasing 1 bit costs $k_B T \ln 2 = 0.693$ nats. Here: acquiring criticality (a binary phase) costs 2.303 nats. The factor difference ($2.303/0.693 \approx 3.32$) reflects the multi-dimensional nature of a criticality transition vs a single-bit erasure.

TENSOR-MATH ANALOGY: criticality_lift is the functor from the ”generic” category of subcritical systems to the ”Φ_c-structured” category where every object has an associated RG fixed point. The cost 2.303 is the ”action” of the functor — the price of the structural promotion.

ASYMMETRY: There is no ”criticality_lower” functor. Once Φ_c is in context, the F-floor ratchet prevents downstream operations from reducing F. This encodes the thermodynamic irreversibility of phase transitions: you cannot un-boil an egg by running the monad backwards.

DEMO: lift on Hv1_human_closed (F_{high}, Φ_{sub}) — should PASS with cost.

LIFT BLOCKED: F gate not met for Hv1_human_closed [BLOCKED] lift(critical) — Φ_c lift not applicable: D_∞ or $G \geq G_\daleth$ required for Φ_c eligibility

CONTRAST: What if F is LOW? (Try on a low-fidelity synthon) conductingelectron (F_{high}) → lift passes (gate: $F \geq F_\hbar$ met) phonon_acoustic (F_{low}) → lift BLOCKED ($F_{\text{low}} < F_\hbar$ required) This correctly encodes: phonons cannot be criticality-lifted; they are not the order parameter. Only high-fidelity recognition modes can undergo a symmetry-breaking transition.

ğ5 PATH — Geodesic in HotSwap Kleisli Category

5.1 path(AtHv1_silent, AtHv1_primed) — discrete topology change

SETUP: AtHv1_silent : $T_\in$ (network; RSN locks topology) AtHv1_primed : $T_\bowtie$ (bowtie; RSN removed, voltage-responsive loop)

HotSwap hard constraint: T must be identical along the path. $T_\in \neq T_\bowtie \rightarrow$ NO PATH in the HotSwap graph.

TENSOR-MATH ANALOGY: In differential geometry: path-connectedness within a homotopy class. $T_\in$ and $T_\bowtie$ are in DIFFERENT homotopy classes of the topology space. There is no continuous deformation from $T_\in$ to $T_\bowtie$ — the transition requires a topological jump (equivalent to tearing the manifold). This is a "1st-order-like" morphism: latent cost > 0 with no intermediate states.

PHYSICAL MEANING: Mechanical priming (membrane stretch) is not a smooth conformational change. It releases the RSN kinetic trap all at once — three primitives (K, T, P) change simultaneously. $d = 3.3$ nats. No smooth path.

PATH BLOCKED: no path from AtHv1_silent to AtHv1_primed Reason: T-class boundary ($T_{\text{network}} \neq T_{\text{bowtie}}$) This confirms: mechanical priming is a discrete 1st-order jump.

5.2 path(2GBI_inhibitor, HIF_inhibitor) — scaffold evolution barrier

SETUP: 2GBI : $T_\in$ (condensed bicyclic network; charge delocalised ring) HIF : $T_|$ (flexible linear linker; two independent pharmacophores)

PATH STATUS: BLOCKED ($T_\in \neq T_|$)

DESIGN CONSEQUENCE: Scaffold optimisation by chemical synthesis (adding substituents, adjusting ring substitution) cannot bridge 2GBI $\rightarrow$ HIF. The evolution from 2GBI-class to HIF-class is a DESIGN DISCONTINUITY: it requires a new synthesis strategy, not incremental SAR. This is why Papers 1$\rightarrow$2 (in the Webster/Tombola series) represent a genuine paradigm shift in Hv1 inhibitor design, not iteration.

TENSOR-MATH ANALOGY: In algebraic K-theory: the "suspension" isomorphism connects $K_0(T_\in\text{-class})$ to $K_0(T_|\text{-class})$ — but only via an explicit stabilisation construction (adding a trivial factor), which corresponds to adding a flexible spacer and rebuilding the pharmacophore. That IS the HIF design.

PATH BLOCKED: $T_\in \rightarrow T_|$ requires T-class redesign. $d(2GBI, HIF) = 1.500$ nats The distance encodes the design effort; the blocked path encodes irreversibility.

5.3 path(topological_insulator, conducting_electron) — topological gap

SETUP: topological_insulator : D_{supra}, $T_\bowtie$, Ω_{Z_2}, Φ_{sub} conducting_electron : D_{mol}, $T_|$, Ω_0, Φ_{sub}

TWO BARRIERS: 1. D: SUPRAMOLECULAR $\neq$ MOLECULAR (bulk TI vs point particle) 2. T: $T_\bowtie \neq T_|$ (Dirac cone topology vs linear dispersion)

PATH STATUS: BLOCKED (two independent class barriers)

ASYMMETRY (SYNTHONICON_LANG.md ğ3e): path(TI $\rightarrow$ Fermi liquid) is blocked by D/T mismatch. path(Fermi liquid $\rightarrow$ TI) is also blocked (reverse). But the COST is asymmetric: TI $\rightarrow$ trivial metal costs $\Delta\xi > 0$ (destroying topological order releases entropy); trivial $\rightarrow$ TI costs MORE (ordering requires investment). The F-floor ratchet encodes this: once in the TI class (F_{high}), the floor is raised and subsequent operations cannot lower F.

PHYSICAL MEANING: Topological protection IS the blocked path. A Majorana edge mode cannot continuously deform to a trivial fermion without closing the bulk gap. The HotSwap path-block is the algebraic encoding of the bulk-boundary correspondence: the gapless surface is protected precisely because there is no smooth path to the trivial phase.

PATH BLOCKED: topological gap is algebraically protected. D-barrier: Dimensionality.SUPRAMOLECULAR $\neq$ Dimensionality.MOLECULAR T-barrier: Topology.CYCLIC_BOWTIE $\neq$ Topology.LINEAR Ω-gap: TopoIndex.Z2_CLASS $\to$ TopoIndex.TRIVIAL

ğ6 PIPELINES — Monadic Composition (SynthonM)

6.1 Hv1 cross-species conservation (from hv1_paper_reproduction.syn)

PIPELINE: start: Hv1_human_open >>= meet(AtHv1_primed) # both $T_{\bowtie}$ — no conflict; Φ_c dominates >>= join(PsHv1_constitutive) # d = 0.000; near-trivial join >>= assert(Φ == Φ_c) # H-bond chain preserved >>= assert(phi_c_score $\geq$ 0.35) # Varma weight for MOLECULAR/LOCAL

MONAD SEMANTICS: Each >>= threads the current synthon through the next operation. MaybeT: if any step returns None (conflict), the rest short-circuit. WriterT: $\Delta\xi$ accumulates — here 0.0 at every step (all within class). StateT: f_{floor} set by join to F_{eth}; criticality_ok unchanged (no lift).

Result : PASS Output : $\langle D_{\wedge}; T_{\bowtie}; R_{\supseteq}; P_{\text{directional}}; F_{\hbar}; K_{\text{mod}}; G_{\sqsupset}; \Gamma_{\text{and}}(\text{SELECTIVE}); \Phi_c\rangle$ $\Delta\xi$ total : 0.000 nats F-floor : $F_{\hbar}$

RESULT: join(meet(Hv1_human_open, AtHv1_primed), PsHv1_constitutive) $\Delta\xi_{\text{CP}}$ = 0.000. Cross-species conservation demonstrated algebraically.

6.2 mplus (<|>) — fallback design strategy

PIPELINE LOGIC: strategy_A: start(Hv1_open) >>= meet(2GBI) # will BLOCK (T-conflict) strategy_B: start(Hv1_open) >>= meet(AtHv1_primed) # will PASS

```
result = strategy_A.mplus(strategy_B)
-> try A; on BLOCK, automatically fall back to B.
```

MONAD SEMANTICS (mplus = MonadPlus operation <|>): mplus is the "choice" combinator in the Maybe monad: Nothing <|> Just x = Just x Here: BLOCKED <|> PASS = PASS Cost of the successful branch accumulates; failed branch cost is lost (WriterT append-only; but MaybeT discards the failed branch output).

THIS IS ALGEBRAICALLY CORRECT: mplus models branching design paths. In retrosynthesis: "try this route; if blocked by a protecting group conflict, try the alternative." The F-floor from the failed branch does NOT transfer (the failed branch never raised the floor).

strategy_A (meet 2GBI) : BLOCKED strategy_B (meet AtHv1) : PASS mplus result : PASS — strategy_B succeeded Output : $\langle D_{\wedge}; T_{\bowtie}; R_{\supseteq}; P_{\text{directional}}; F_{\hbar}; K_{\text{mod}}; G_{\sqsupset}; \Gamma_{\text{and}}(\text{SELECTIVE}); \Phi_c\rangle$

6.3 Quantum co-assembly pipeline: Cooper pair $\otimes TIsurface$

PIPELINE: start: cooper_pair (Ω_Z, Φ_c, $T_{\bowtie}$) >>= tensor(topological_insulator, $\lambda = 0.3$) >>= assert(criticality_phase == Φ_c)

PHYSICAL MEANING: Proximity effect: a Cooper pair condensate adjacent to a TI surface can induce topological superconductivity. The tensor product predicts the primitive structure of this co-assembly. The assert checks whether Φ_c is preserved in the heterostructure.

Ω in tensor: $\max(\Omega_Z, \Omega_{Z_2}) = \Omega_Z$ ($\mathbb{Z}$-class $>$ $\mathbb{Z}_2$-class in protection ordinal) T: $T_{\bowtie} \otimes T_{\bowtie} \to T_{\bowtie}$ (bowtie preserved; Dirac + pairing loop) Φ: $\Phi_c \otimes \Phi_{\mathrm{sub}} \to \Phi_c$ (criticality propagates)

This tensor product is the algebraic encoding of the proximity effect that generates topological superconductor phases (class D/DIII).

Result: PASS Output : $\langle D_{\triangle}; T_{\bowtie}; R_{\supseteq}; P_{\mathrm{pm\_pseudo}}; F_{\hbar}; K_{\mathrm{slow}}; G_{\gimel}; \Gamma_{\mathrm{and}}(\mathrm{SELECTIVE}); \Phi_c \rangle$ $\Delta\xi_{\mathrm{CP}}$: 13.854 nats Ω : None (higher Ω propagates in tensor) Φ : CriticalityPhase.CRITICAL (Φ_c propagates)

ğ7 DECOMPOSITIONS — Factor · Cofactor · Kernel · Principal · Project · Complement

7.1 cofactor(cooper_pair, conducting_electron) — inverse tensor: what is the pairing partner?

SETUP: cooper_pair : $\langle D_{\mathrm{mol}}; T_{\bowtie}; R_{\supseteq}; P_{\mathrm{sym}}; F_{\hbar}; K_{\mathrm{slow}}; G_{\gimel}; \Gamma_{\wedge}(\mathrm{SPEC}); \Phi_c; \Omega_Z \rangle$ conducting_electron: $\langle D_{\mathrm{mol}}; T_{|}; R_{\supseteq}; P_{\mathrm{donor}}; F_{\hbar}; K_{\mathrm{fast}}; G_{\beth}; \Gamma_{\wedge}(\mathrm{SEL}) \rangle$

QUESTION: tensor$(electron, ?) \approx$ cooper_pair Find the "pairing partner" — the quasiparticle that, when tensored with the conducting electron, produces the Cooper pair primitive tuple.

COFACTOR RULES (inverting tensor per axis): F (min-dominant): tensor$[F] = \min(A_F, B_F)$ $A_F = F_{\hbar}$, $C_F = F_{\hbar} \to A$ explains F; $B_F = F_{\hbar}$ (must be equally high) K (min-dominant): tensor$[K] = \min(A_K, B_K)$ $A_K = K_{\mathrm{fast}}$, $C_K = K_{\mathrm{slow}} \to A$ is NOT the bottleneck; $B_K = K_{\mathrm{slow}}$ (B is the slow partner) G (max-dominant): tensor$[G] = \max(A_G, B_G)$ $A_G = G_{\mathrm{local}}$, $C_G = G_{\mathrm{meso}} \to B_G = G_{\mathrm{meso}}$ (B contributes the coherence length) T (promotion): $C_T = T_{\bowtie}$, $A_T = T_{|} \to B$ must contribute $T_{\bowtie}$ (B has the pairing loop topology) Φ (join-dominant): $C_\Phi = \Phi_c$, $A_\Phi = \Phi_{\mathrm{sub}} \to B_\Phi = \Phi_c$ (B carries the criticality) Ω (join-dominant): $C_\Omega = \Omega_Z$, $A_\Omega = \Omega_{\mathrm{trivial}} \to B_\Omega = \Omega_Z$ (B carries the topological invariant)

PHYSICAL MEANING: The inferred partner B has: K_{slow} (the condensate timescale, not the Fermi velocity), $G_{\mathrm{mesoscale}}$ (the coherence length), $T_{\bowtie}$ (the pairing loop), Φ_c (the superfluid phase transition), Ω_Z (the winding number). This is the PHONON-DRESSED ELECTRON — the retarded interaction that makes the other electron "look slow" through phonon exchange. The cofactor correctly reconstructs the effective retarded partner.

STATUS : PASS RESULT : $\langle D_{\wedge}; T_{\bowtie}; R_{\supseteq}; P_{\mathrm{pm\_sym}}; F_{\hbar}; K_{\mathrm{slow}}; G_{\gimel}; \Gamma_{\mathrm{and}}(\mathrm{SPECIFIC}); \Phi_c; \Omega_Z \rangle$

PER-AXIS ROLES: F : BOTTLENECK A is fidelity bottleneck; $B \geq$ Fidelity.HIGH K : BOTTLENECK B sets K floor = KineticCharacter.SLOW G : CONTRIBUTOR B is the G=Granularity.MESOSCALE contributor D : EXPLAINED A explains D; B needs only $D_{\mathrm{MOLECULAR}}$ Φ : CONTRIBUTOR B must carry Φ_c (A doesn't have it) Ω: CONTRIBUTOR B carries Ω=TopoIndex.Z_CLASS T : PASSTHROUGH B contributes T=Topology.CYCLIC_BOWTIE (A has Topology.LINEAR) R : EXPLAINED A explains R=RecognitionMode.NON_COVALENT P : PASSTHROUGH B contributes P=Polarity.SELF_COMPLEMENTARY_SYM (A has Polarity.DONOR)

Γ: PASSTHROUGH B contributes Γ=InteractionGrammar.SPECIFIC_AND (A has InteractionGrammar.SELECTIVE_AND)

Φ_c SOURCE: cofactor $\implies$ Cofactor(cooper_pair | conducting_electron) computed successfully $\implies$ F: BOTTLENECK — A is fidelity bottleneck; $B \geq$ Fidelity.HIGH $\implies$ K: BOTTLENECK — B sets K floor = KineticCharacter.SLOW $\implies$ G: CONTRIBUTOR — B is the G=Granularity.MESOSCALE contributor $\implies$ D: EXPLAINED — A explains D; B needs only $D_{\text{MOLECULAR}}$ $\implies$ Φ: CONTRIBUTOR — B must carry Φ_c (A doesn't have it) $\implies$ Ω: CONTRIBUTOR — B carries Ω=TopoIndex.Z_CLASS $\implies$ T: PASSTHROUGH — B contributes T=Topology.CYCLIC_BOWTIE (A has Topology.LINEAR) $\implies$ R: EXPLAINED — A explains R=RecognitionMode.NON_COVALENT $\implies$ P: PASSTHROUGH — B contributes P=Polarity.SELF_COMPLEMENTARY_SYM (A has Polarity.DONOR) $\implies$ Γ: PASSTHROUGH — B contributes Γ=InteractionGrammar.SPECIFI (A has InteractionGrammar.SELECTIVE_AND)

7.2 principal_decomp(GNF2) — join-irreducible basis decomposition (SVD analog)

SETUP: GNF-2 : $\langle D_\wedge; T_{\text{branched}}; R_\supseteq; P_{+-}; F_\varepsilon; K_{\text{mod}}; G_\daleth; \Gamma_\to(\text{SEL}); \Phi_c; \Omega_0\rangle$

The decomposition produces an ordered list of join-irreducible factors. Each factor = single primitive contribution above the constraint-bottom. Reading order = most constraining $\to$ least constraining.

TENSOR-MATH ANALOGY: In a product lattice $L = L_F \times L_K \times L_G \times L_{\text{categorical}}$, every element s has a unique Birkhoff representation as a join of join-irreducible elements. This is the lattice-theoretic analog of expressing a vector in terms of basis vectors. The "principal" decomposition orders these by their ξ_{CP} contribution — the analog of ordering singular values by magnitude.

EXPECTED FACTORS (rough order): 1. Φ_c component — allosteric criticality (hardest to satisfy) 2. G_{meso} component — mesoscale propagation 3. K_{mod} component — kinetic character 4. F_{med} component — fidelity floor 5. Categorical skeleton (D, T, R, P, Γ unchanged)

DESIGN USE: tells you which primitive of GNF-2 is the HARDEST to engineer. If you're trying to improve GNF-2, start with factor 1 (highest ξ contribution).

FACTORS (4 total, each is a join-irreducible atom): [1] atom[$F = F_{\text{eth}}$] non-bottom: $F = F_{\text{eth}}$ [2] atom[$K = K_{\text{mod}}$] non-bottom: $K = K_{\text{mod}}$ [3] atom[$G = G_\daleth$] non-bottom: $G = G_\daleth$ [4] skeleton(GNF2_allosteric) non-bottom: Φ_c

ξ BALANCE: 0.000 nats $\implies$ Factor 1: F contribution = Fidelity.MEDIUM $\implies$ Factor 2: K contribution = KineticCharacter.MODERATE $\implies$ Factor 3: G contribution = Granularity.MESOSCALE $\implies$ Skeleton: categorical primitives of GNF2_allosteric

7.3 project + complement_rel — Heyting pseudocomplement (complementary design)

STEP 1: project(cooper_pair, ["criticality_phase", "topo_index"]) Retain only Φ and Ω; zero out all other primitives. Analogous to: $\pi_i(v)$ — project vector v onto the $\{i\}$-th coordinate subspace.

PROJECTED: $\langle D_\wedge; T_{\text{linear}}; R_\supseteq; P_{\text{pm\_sym}}; F_\ell; K_{\text{fast}}; G_\sqsupset; \Gamma_{\text{or}}(\text{BROAD}); \Phi_c; \Omega_Z\rangle$ RETAINED : Φ = CriticalityPhase.CRITICAL, Ω = TopoIndex.Z_CLASS ZEROED : D, T, R, P, F, K, G, Γ

STEP 2: complement_rel(gnf2, context=projected, target=cooper_pair) Find the maximal $x \leq$ GNF-2 such that: (1) $x \sqcap \text{projected} = \perp$ (x has no overlap with the Φ/Ω projection) (2) $x \sqcup \text{projected} \geq \text{cooper\_pair}$ (together they cover the target)

TENSOR-MATH ANALOGY: In a Heyting algebra (the algebraic model of intuitionistic logic): $a \Rightarrow b = \bigvee\{x : x \wedge a \leq b\}$ (the IMPLICATION / pseudocomplement) Here: complement_rel(GNF-2, context, target) = GNF-2's "contribution" to reaching the cooper_pair target AFTER accounting for what context already covers.

```
This is also the constructive analog of the QUOTIENT in module
    theory:
GNF-2 / context = the part of GNF-2 that context does NOT
   already explain.
```

DESIGN MEANING: The result tells you: "if the Φ/Ω structure is already given by the topological material, what does the molecular GNF-2 uniquely contribute toward the cooper_pair target?" Answer: $K_{\text{slow}} + G_{\text{meso}} + T_{\text{branched}}$ (the slow, mesoscale, branched kinetics that the topological material alone cannot provide).

SATISFIED : False $\implies$ K: complement value = KineticCharacter.MBL $\implies$ G: complement value = Granularity.MESOSCALE $\implies$ F: cannot satisfy target F=Fidelity.HIGH $\implies$ F: complement value = Fidelity.MEDIUM $\implies$ D: context covers target; $x[D]$ = Dimensionality.MOLECULAR $\implies$ T: $x[T]$ = Topology.CYCLIC_BOWTIE (complement of context Topology.LINEAR) $\implies$ R: context covers target; $x[R]$ = RecognitionMode.NON_CO $\implies$ P: context covers target; $x[P]$ = Polarity.DONOR_ACCEPTOR $\implies$ Γ: $x[\Gamma]$ = InteractionGrammar.SPECIFIC_AND (complement of context InteractionGrammar.BROAD_OR) $\implies$ Φ: context covers target; $x[\Phi]$ = None $\implies$ Ω: context covers target; $x[\Omega]$ = TopoIndex.TRIVIAL

Run individual sections: python TENSOR_OPS_DEMO.py –section tensor

All mathematical symbols are now wrapped in $ (or

$fordisplayedequations)asrequested.Thechangesensureconsistentrenderingandmaintaintheoriginalc$